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Highly robust soft-rigid connections via mechanical interlocking for assembling ultra-stretchable displays

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Stretchable electronic circuits can seamlessly conform to irregular and dynamic surfaces with high integration. However, current stretchable configurations typically have limited stretchability due to the lack of robust connections between soft interconnects and rigid electronics. Here, we printed highly stretchable metal–polymer conductors on thermoplastic elastomers as interconnects. We developed electronic vests with porous surfaces for rigid electronics and introduced polyester hot-melt adhesives to strengthen connections between soft interconnects and rigid electronics. After thermal bonding, the adhesive penetrates the porous surface of electronic vests, creating a mechanical interlock and providing an adhesion force of 8.34 N/cm for the connection (3× higher than conductive adhesives). Thus, rigid electronics of different sizes and different pin counts can form strong connections to soft interconnects, achieving a maximum strain tolerance of ~700% (10× higher than conductive adhesives). We achieved highly integrated ultra-stretchable displays that can withstand stretching up to 220% without dead pixels.

Stretchable electronics can seamlessly integrate with highly dynamic and irregular surfaces. They have great potential in a broad range of emerging fields, including implantable electronics[1–3], soft robotics[4,5], and human–machine interfaces[6–8]. Recently, we have witnessed the encouraging development of stretchable conductors realized by novel materials[9–14] and structures[15–18]. Some of these reported conductors have excellent electrical conductivity close to that of silver and amazing stretchability of up to 1000%[19,20]. However, stretchable circuits based on these conductors have not performed satisfactorily and usually suffer from poor stretchability limited by connections between stretchable conductors and rigid electronics (soft–rigid connections). To achieve fully stretchable circuits based on stretchable conductors, it is inevitable that stretchable conductors must be connected with commercially available rigid electronic devices [21–25], such as microcontrollers, batteries, and amplifiers, because soft substitutes for most of these electronics have not yet been developed. Although solder pastes can be used to solder rigid electronics directly to wavy-structured Cu/Au interconnects and achieve a stretchability of about 50% [26,27], solder paste is not suitable for soldering composite stretchable conductors due to the influence of wettability. To obtain robust soft–rigid connections of composite stretchable conductors, conductive adhesives such as silver pastes[28,29], carbon nanomaterials[30,31], and liquid metal composites[23,32] are used to reinforce the connections. However, these conductive adhesives have limited adhesion, when the stretchable circuit deforms, mechanical/electrical failures usually occur at the soft–rigid connections due to stress concentration. Liquid metals are suitable for connecting rigid electronics with large pins, such as resistors, capacitors, and LEDs. However, they can cause short circuits when used to solder microcontrollers with dozens of small pins due to their fluidity[33]. Some vapors can also be used to strengthen the soft–rigid connections[25,34], but damage from toxic vapors needs to be avoided during preparation.

Here, we report a thermal bonding strategy for assembling ultra-stretchable circuits with robust soft–rigid connections. We solder rigid electronics onto the electronic vests (E-vests) with porous surfaces that can provide electrical and mechanical connections to soft circuits. We use polyester hot-melt adhesive film (PES HMA) as the reinforcement film that could penetrate into the pores and cavities of the surface of e-vests to form a mechanical interlock after thermal bonding (pressure and heat were applied), thus greatly enhancing the adhesion between soft–rigid connections, enabling a maximum strain tolerance of ~700% for rigid electronics in elastic substrates. The substrate of the circuit is composed of a thermoplastic elastomer, thus the assembly of highly stretchable circuits can also be achieved through thermal bonding. The thermal bonding strategy does not

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involve expensive nanofabrication processes and toxic chemicals, which is suitable for large-scale roll-to-roll production. Based on the thermal bonding strategy, we achieved a highly integrated ultra-stretchable display (containing 148 rigid electronics) that can withstand a maximum stretch of 220% without dead pixels. After 2000 stretching cycles at 100% strain, the display had a defective pixel rate of <5%.

Results and discussion

Design of the soft–rigid connection for stretchable display

Figure 1a shows the structure and the assembly of the stretchable display. We use thermoplastic elastomer polyurethane (TPU) as the substrate and encapsulation for stretchable displays. We use screen-printing technology to print metal–polymer conductors (MPCs) on TPU substrates as interconnects for stretchable displays. MPCs are highly stretchable conductors composed of liquid metal and polyvinylpyrrolidone. In our previous work, MPCs have demonstrated excellent conductivity, stretchability (380,000 S/m at 1000% strain), and repeatability ($\Delta R/R < 3\%$ after 10,000 cycles), which means that MPC wires can operate for long periods of time without failure even under extreme deformation[35,36]. Although the average strain in most stretchable circuits does not exceed 100%, it is necessary for MPC to withstand strains up to 1000%. This is because there is a strain concentration at the interface of the soft–rigid connection, where the strain will be much larger than the average strain. Thus, the excellent stretchability of the MPC can ensure that the MPC wires will not fail in the stress concentration area.

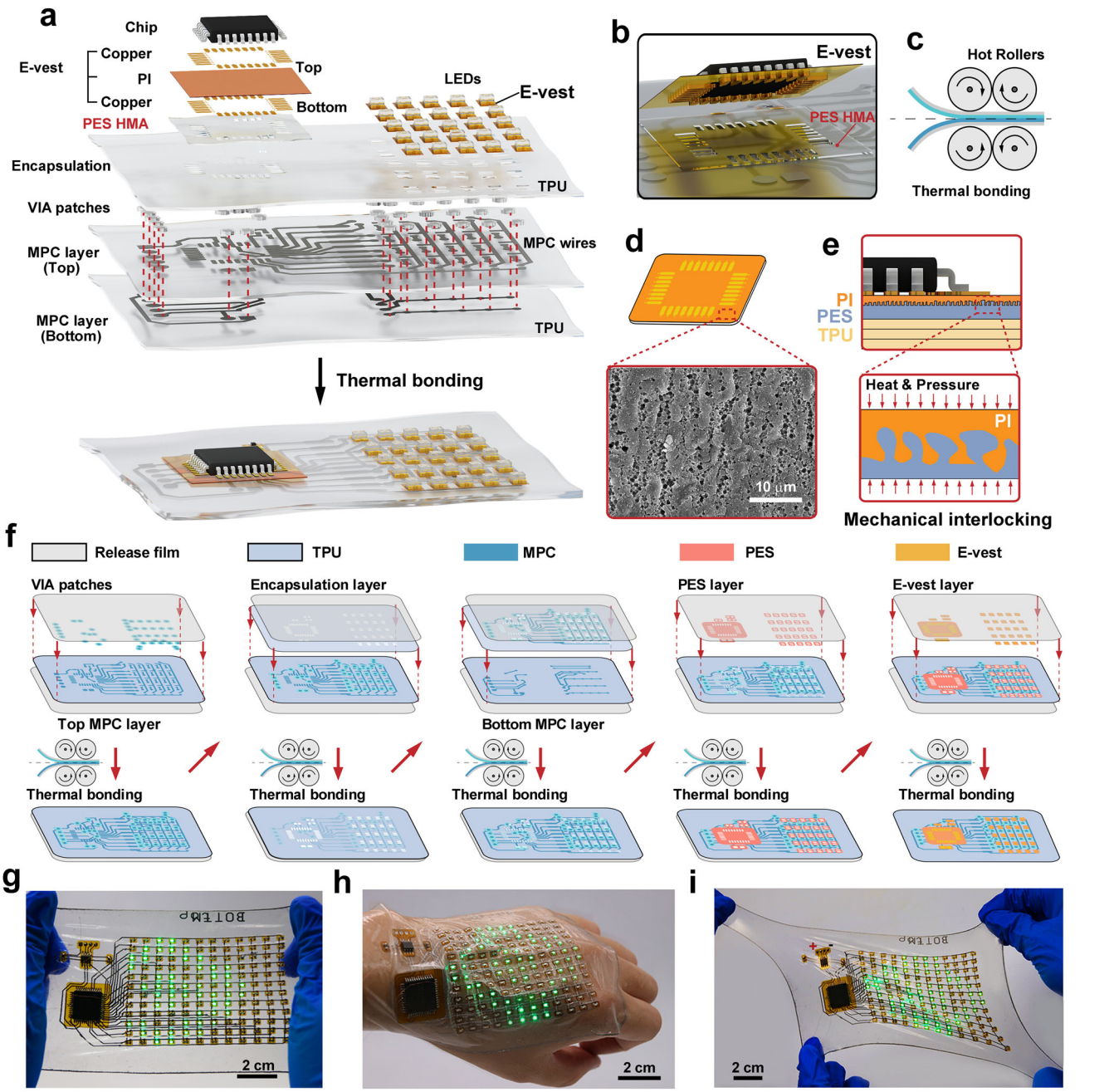


Fig. 1 | Schematic illustrations and optical images of the stretchable display. **a** Exploded view of a stretchable display and assembly diagram of the display. **b** Structure of the soft–rigid connection. **c** Schematic illustrations of thermal bonding. **d** E-vest composed of porous PI. **e** Mechanism of the soft–rigid connections. **f** Fabrication process of the stretchable display. **g** The Optical image of a stretchable display with scrolling messages. **h** The stretchable display is conformal to the skin. **i** The 12 × 12 LED display under stretching. The LED display is powered by two thin enameled wires connected to a 5 V power supply.

The vertical interconnect accesses (VIA) patches are circular TPU sheets printed with a circular MPC pattern, which are responsible for the electrical connection of different layers. The VIA patches are aligned with the holes on the top MPC layer and the wires on the bottom MPC layer. After thermal bonding (Fig. 1c), the TPU on the VIA patch will fuse with the TPU on the top MPC layer and form a mechanical connection. MPC on the VIA patch will be in direct contact with the MPC wires on the top and bottom layers (Supplementary Fig. 1).

We create e-vests for rigid electronic components that can connect stretchable substrates to the rigid chip. The e-vests can provide electrical and mechanical connections for rigid electronics on the stretchable substrate (Fig. 1b). The e-vests are customizable commercially available flexible printed circuits (FPC) composed of porous polyimide (PI) and Cu films with identical patterns on both sides. The Cu films on the top and bottom layer of the PI are interconnected via plated-through holes (Supplementary Fig. 2a). The copper trace on the top layer of the PI can directly solder to the pins of a chip, while the copper trace on the bottom layer of the PI can directly contact the MPC wires and form electrical connections. The electrical connections are located inside the e-vest and are almost not subjected to stress during stretching. The mechanical connections of the e-vest are located at the edges of the electronic vest and can be greatly reinforced by the mechanical interlocking (Fig. 1e). As long as the edge of the electronic vest does not detach from the substrate, the electrical connection inside the electronic vest can remain stable.

Mechanical interlocking is achieved by the penetration of thermoplastic polymers into the porous surface of the e-vest when pressure and heat are applied (Fig. 1e). We adopted the reinforcement film PES HMA to further strengthen the mechanical connection between the e-vest and the substrate. Through thermal bonding (Fig. 1c), the different layers and components in the stretchable display can be firmly assembled due to the fusion of thermoplastic polymers (TPU and PES) with different surfaces. To assemble the stretchable display, we first independently fabricate each layer and components of the stretchable display, including two MPC interconnects printed TPU layers, VIA patches, reinforcement films, and e-vest (Fig. 1a). We then use a laminator to thermally bond those components layer by layer to obtain the soft printed circuit board (Fig. 1f). Finally, microcontrollers, resistors, and LEDs can be soldered directly to e-vest through solder paste to be a stretchable display. LEDs in the stretchable display could be controlled individually to create a scrolling message. Although the stretchable display (Fig. 1g) contains a large number of rigid electronics of different sizes and varying pin counts, it exhibits excellent conformability (Fig. 1h) and stretchability (Fig. 1i).

Characterization of the soft–rigid connection

The excellent stretchability of our display originates from the robust soft–rigid connection enabled by the e-vests and reinforcement film. Rigid electronics typically have poor adhesion and small contact areas to stretchable substrates. Here, we developed e-vests for rigid electronics to increase adhesion and contact areas between rigid electronics and soft substrates (Fig. 1b). The e-vests are customizable commercially available FPC, and Supplementary Fig. 2b presents the structure of a soft–rigid connection enabled by e-vest. The e-vests are composed of polyimide (PI) and copper film, which can be connected to a soft substrate by thermal bonding. Finally, the rigid electronics can be soldered to the e-vests. We tested Young's modulus of TPU, PES HMA, and PI. Young's modulus of TPU is 3.3 MPa, Young's modulus of PES HMA is 270 MPa, and Young's modulus of PI is about 2280 MPa. The pins of the chip are usually made of tinned steel, whose Young's modulus is about 200 GPa.

Two types of interfaces are created after the thermal bonding in stretchable circuits (Fig. 1e): the interface between thermoplastic polymers (such as TPU–TPU and TPU–PES interfaces) and the interface between the thermoplastic polymer and PI on the E-vest (such as PES–PI or TPU–PI interfaces). The interfaces between thermoplastic polymers often have high bonding strength after thermal bonding due to the inter-diffusion of the polymer chains at the bonding interface[37]. Tensile tests of thermally bonded two TPU strips show that the junction of the two TPU strips did not separate even when the TPU broke at about 1200% strain (Supplementary Fig. 3a), which suggests that the interface between thermoplastic polymers is not the limiting factor for stretchability in circuits.

The limiting factor for stretchability originates from the interface between the thermoplastic polymer and PI, where the bond strength is relatively low due to mechanical interlocking.

We adopt two strategies to enhance the mechanical interlocking effect. Firstly, we can increase the number of pores and cavities on the PI substrate of the e-vests. The thermoplastic polymer will melt in the thermal bonding, wet the PI substrate, and penetrate pores, cavities, and irregularities on the surface. After solidification, the thermoplastic polymer acts like multiple hooks anchored on the PI surface (Fig. 1e). We tested the adhesion force between the TPU and different e-vests composed of smooth PI (sPI) and porous PI (pPI), respectively (Supplementary Fig. 2c). The e-vests can directly connect to the TPU surface through thermal bonding. The adhesion force between the sPI and the TPU is about 1.13 N/cm, while the adhesion force between the pPI and the TPU is about 3.04 N/cm, which indicates that the e-vest composed of pPI provides more cavities for the penetration of molten TPU. Tensile tests of thermally bonded TPU and different PI strips show that the junction will gradually detach during stretching and finally separate at a strain of 117% ± 37% (sPI) and 341% ± 62% (pPI) due to insufficient adhesion (Supplementary Fig. 3b). When stretched, detachment of the e-vests from the circuit is the leading cause of failure in stretchable circuits.

Secondly, we can increase the wettability of the penetrated polymer in the molten state. In order to further increase the effect of mechanical interlocking, we can select a hot-melt adhesive with better wettability with the PI substrate as a reinforcement film, thus increasing the bonding strength between e-vest and TPU. We screened the commercial hot-melt adhesive films such as PES, TPU, polyolefin (PO), ethylene-vinyl acetate copolymer (EVA), and polyamide (PA) to reinforce the soft–rigid connection (Fig. 2b). PES can significantly improve the adhesion force (8.34 N/cm) between e-vest composed of pPI and TPU. The PES has the smallest contact angle with PI after melting (Fig. 2c and Supplementary Fig. 4), so the molten PES can easily wet the cavities on the PI surface, resulting in stronger mechanical interlocking.

We also compared the PES with commonly used conductive adhesives for connecting electronics, such as Ag paste and 3M 9705. (3M 9705 is a pressure-sensitive adhesive with anisotropic electrical conductivity, which allows electrical conduction in the z-direction.) We have found that the connection strength achieved by thermal bonding is 3–4 times stronger than conductive adhesives (Fig. 2b).

To find out how the thickness of the substrate affects stretchability, elastic TPU with different thicknesses is thermally bonded with PI through PES to simulate the soft–rigid connection during stretching (Fig. 2d). We found the failure of all samples is due to the detachment of PI from PES due to insufficient adhesion. As the thickness of the substrate increases, the soft–rigid connections detach under smaller tensile strain (Fig. 2d). Finite element analysis was performed on an e-vest on substrates of different thicknesses (100, 150, 200 μm) subjected to an average strain of 50%. When stretched, the shear stress on the edges of the electronic vest is the main reason for the failure of the e-vest. It can be seen from Supplementary Fig. 5a that the shear stress is concentrated on the periphery of the e-vest, and the maximum shear stress applied to the edge of the e-vest increases with the increase of the substrate thickness (Supplementary Fig. 5b). Thus, we need to adopt elastic substrates with smaller thicknesses and Young's modulus to obtain a circuit with excellent stretchability.

In order to understand how Young's modulus of the substrate affects the stretchability of the soft–rigid connection, we selected two TPUs with different Young's moduli (3.3 and 11.7 MPa) as the substrates in Fig. 2d (left). Also, the failure of all samples is due to the detachment of PI from PES due to insufficient adhesion. As Young's modulus of the substrate increases, the stress concentration at the periphery of the e-vest becomes greater, so that the adhesion provided by the mechanical interlocking is exceeded at a

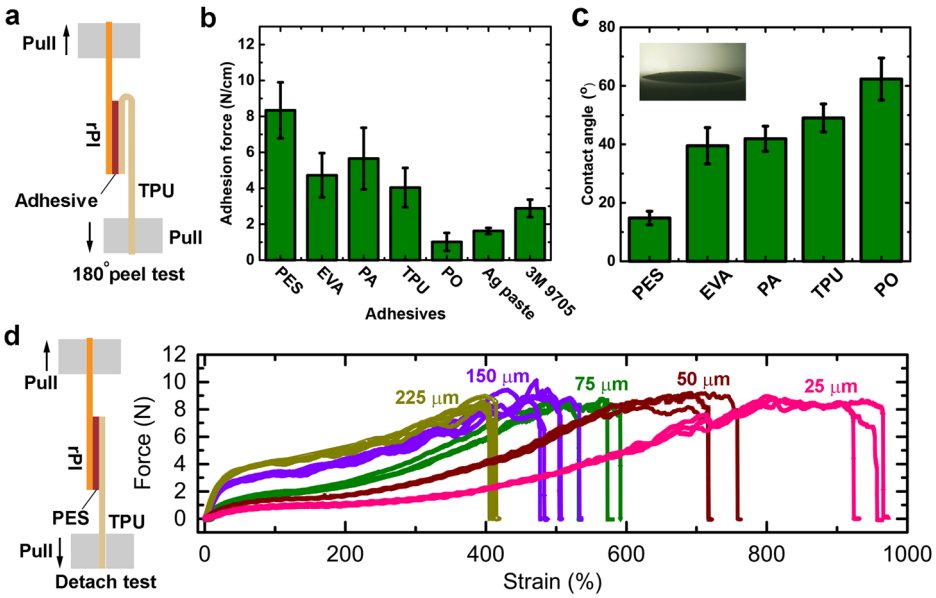


Fig. 2 | Mechanical characterization of the soft–rigid connection. a Schematic illustrations of 180° peel test for measuring the adhesion force. **b** The adhesion force between PI and TPU is reinforced by different materials. The sample size was $n = 7$ for each group. The data are presented as the means ± SD. **c** Contact angles of different materials on PI after melting. **d** The force–strain curve for the PI–TPU connection is reinforced by PES. Curves of different colors represent different thicknesses of TPU.

smaller strain, and the soft–rigid connection detaches at a smaller tensile strain (Supplementary Fig. 6).

The temperature for the thermal bonding also affects the adhesion force between the PES and the PI. Bonding between PI and PES requires a temperature of about 140 °C (Supplementary Fig. 7a). When the temperature exceeds 160 °C, defects such as pores and cavities will appear on the surface of TPU (Supplementary Fig. 7b), thereby reducing the adhesion force.

The e-vests can adhere to the TPU substrate through PES and serve as soft–rigid connections. When the circuit gets stretched, the e-vest can form reliable mechanical and electrical connections with interconnects on the substrates. We carried out tensile tests on 2-pin electronics to demonstrate the high stretchability of the soft–rigid connection (Fig. 3a). To spotlight the superior strain-resistant electrical and mechanical properties of the e-vest, we used different strategies, including e-vest, silver pastes, and 3M 9705 to connect a 100-ohm resistor to the MPC wires. We stretched these circuits, each enabled by different connections to the electrical failure. As a control, we also stretched the MPC wires without e-vest. In Fig. 3c, the MPC wires with an e-vest connection lost their electrical conductivity at about 720% strain due to e-vest disconnection, while the MPC wires without e-vest remained conductive at up to 900% strain. In contrast, when using 3M 9705 as a connection, the rigid resistor will detach from the substrate and lose electrical connection at about 40% strain. When using the Ag paste as the connection, the silver solder will break and detach from the substrate at a strain of about 150% strain. Figure 3c shows the circuit with LED under different strains.

As shown in the FEA simulation (Fig. 3b), the electrical connections of the e-vest remain in a low deformation state during stretching because the electrical connections are inside the PI substrate with high Young's modulus. The electrical connections of the e-vest are almost unstressed during extreme deformation, which is why the stretchable circuits based on the electronic vest can remain stable. The mechanical connection of the electronic vest is located at the edge of the electronic vest. Although there is strain concentration around the connection, the mechanical connection strength is greatly enhanced through mechanical interlocking, and the connection can still be maintained under large deformation.

We analyzed the failure process of e-vests. The detachment of e-vest from the substrate does not happen immediately but is a process in which the degree of detachment gradually accumulates as the strain grows (Supplementary Fig. 8). When the strain reaches about 530%, the corners of the e-vest start to detach (Fig. 3e). When the strain is >530%, the detachment gradually propagates inward. When the strain reaches about 720%, the copper electrical contacts on one side of the E-vest are completely separated from the MPC on the substrate, eventually leading to the e-vest's failure. Figure 3d shows that the LED connected to the MPC still lights up under different strains (maximum 700%).

We carried out the cycle tests for the soft–rigid connections enabled by the e-vests. We stretched the MPC wire with the connection of e-vest (100-ohm resistor) from its original length to a strain of 100%, 200%, and 300%, respectively, and then released it to the original length. After 2000 stretch and release cycles, the MPC wires with the e-vest connection maintain their conductivity. When the strain in the stretch cycle is <200%, the e-vest connection shows excellent repeatability (Fig. 3f), and the resistance fluctuation at the maximum strain is <2%. However, when the cyclic strain reaches 300%, the resistance of the MPC wire with the e-vest connection increases with the cycles. We believe the resistance increase is caused by the excessive deformation of MPC, which will cause rapid MPC oxidation. Because the surface area of the liquid metal changes greatly in each stretching cycle, the oxide film on the surface of the liquid metal is constantly broken, and the fresh liquid metal is exposed to the air and oxidizes violently, so the electrical properties of the liquid metal conductors are greatly degraded. E-vests have excellent performance when used to connect multiple electronics. Figure 4a shows that multiple LEDs connected on a soft substrate enabled by e-vests can emit light normally, even under 300% deformation.

The soft–rigid connections enabled by e-vest are also suitable for microcontroller chips with dense pins (Fig. 4b). The chip's pin size and spacing can be enlarged to any size by e-vest to connect with soft conductors (Supplementary Fig. 9). The e-vest can provide both mechanical and electrical connections with the substrate. When there is no mechanical connection between the chip and the circuit, the chip moves when the substrate deforms, causing the chip to lose the electrical connection to the circuit (Fig. 4c). We compared different strategies for soft–rigid connections, such as adopting conductive silver paste, 3M 9705, and bulky liquid metal solders (Fig. 4c). Liquid metal solders are the bulky liquid metals that we apply to the pins with a brush (Supplementary Fig. 10a). For conductive silver paste and 3M 9705, the acceptable tensile strain is generally not more than 200%. By contrast, with the e-vest, whether a 2-pin resistor or a multi-pin chip, the soft–rigid connections can withstand stretchability exceeding 700%. Thus, compared with silver paste and 3M 9705, e-vest can form a larger contact area and greater adhesion with the substrate. Compared with bulk liquid metal solder, the e-vest can effectively avoid short circuits caused by the flow

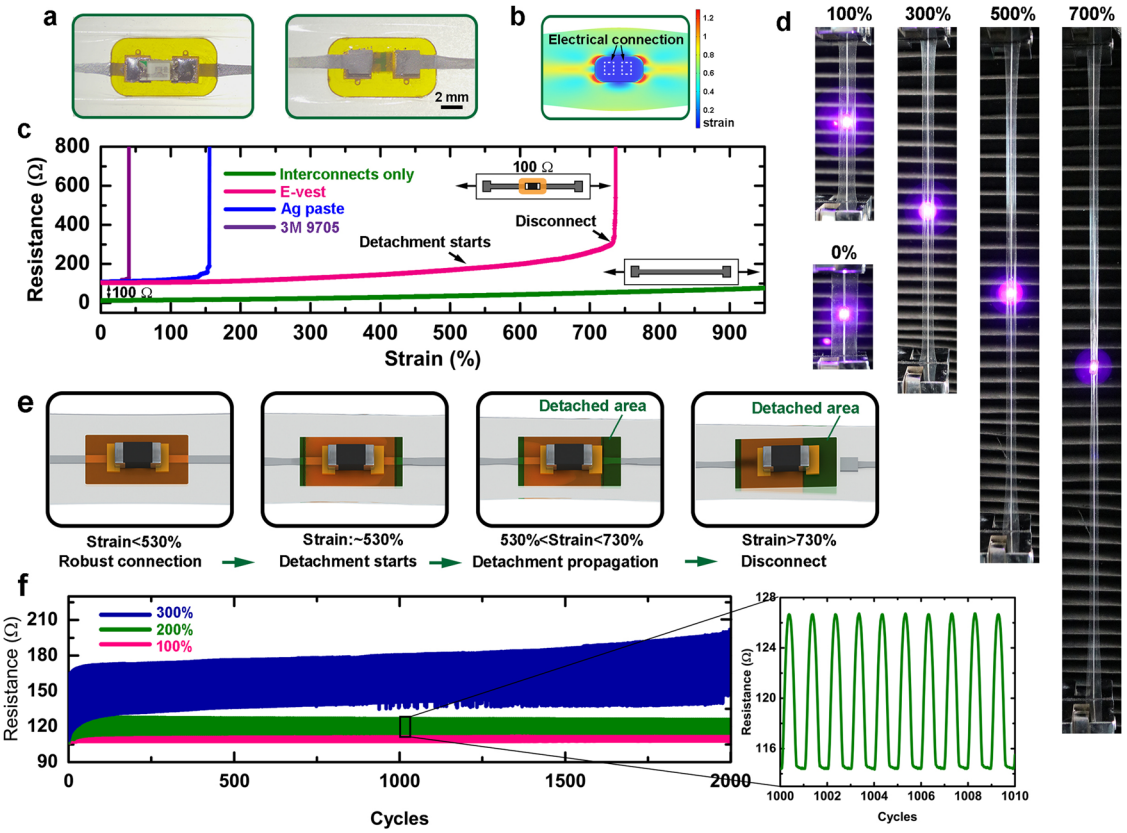


Fig. 3 | Electro-mechanical characterization of the soft–rigid connection. **a** Optical images of the front and back sides of an LED circuit under strain. **b** FEA result of the strain distribution on the circuit based on e-vest. **c** Resistance of the soft–rigid connections enabled by different strategies versus the strain. The green curve represents the resistance of the MPC wire versus the strain. **d** LED circuit under different strain conditions. **e** Schematic showing the detachment process of the E-vest from the substrate under different strains. **f** Real-time monitoring of the resistance of the soft–rigid connection by stretching it from a strain of 0–100%, 200%, and 300% for 2000 cycles.

of bulk liquid metal under stretching. Though a 2-pin resistor circuit implemented with liquid metal solder can achieve up to 900% stretchability (Fig. 4e), liquid metal solders are unsuitable for connecting multi-pin chips because liquid metal cannot form a stable mechanical connection with the chip. When the stretchable circuit deforms, the chip moves, and bulky liquid metal solders on adjacent chip pins will contact and create a short circuit (Supplementary Fig. 10b).

We further characterize how disconnection occurs in multi-pin configurations under strain. The multi-pin e-vest is disconnected in a similar manner to the 2-pin e-vest. When the strain reaches about 490%, the edge of the e-vest in the stretching starts to detach. When the strain is >490%, the detachment gradually propagates inward. When the strain reaches about 700%, the pins of the e-vest in the stretching direction are completely detached from the MPC on the substrate, eventually leading to the e-vest's failure (Fig. 4d).

To demonstrate that the e-vest can be universally utilized for various stretchable devices, we use different elastomers, including TPU, EVA (poly (ethylene-vinyl acetate)), PDMS, and Ecoflex 0030 as the substrate of the stretchable circuits. We printed MPC on TPU, EVA, PDMS, and Ecoflex, respectively, and we used the e-vest to connect a 100-ohm resistor to the MPC wires. We found the failure strain (the strain at which complete electrical failure occurs) of the TPU and EVA both exceed 500%, while the failure strain of PDMS and Ecoflex smaller than 10% (Fig. 4f). We believe that both TPU and EVA are thermal plastic elastomers. During the thermal bonding process, the thermoplastic polymer will melt and fuse with the PES layer through inter-diffusion at the bonding interface. However, PDMS and Ecoflex are silicones with inert surfaces. After high-temperature hot pressing, the PES and silicone surfaces cannot form an effective bond. Thus, in the case of slight deformation, the interface between silicone and PES will detach, resulting in electrical failure.

Different electronic components vary greatly in size. We explored the effect of e-vests with different widths on stretchability. In the tensile test of e-vest with different widths (5, 10, 15, 20 mm), we found that all e-vest can reach a strain of 700% and maintain electrical connection, which suggests that the width of the e-vest has little influence on stretchability (Fig. 4g).

The size of electronic components, especially chips, is usually determined by their packaging. For the convenience of human use, the size of the package of commercial electronic components is usually much larger than the size of the bare die (chip without packaging, Supplementary Fig. 11). The e-vests in this work are designed for commercial electronic components already have a relative large packaging. So it is indeed hard to further reduce the size when designing e-vests for commercially available electronic components. In the future, in order to achieve the miniaturization of electronic components in stretchable circuits, we are going to directly integrate the bare die on the e-vest and develop a lightweight and reliable package. Thus, we can further improve the stretchability and reduce the size of our stretchable circuits based on the e-vests.

The thickness of the circuits will also influence the stretchability of our circuits. A stretchable circuit contains at least two layers of TPU, including the MPC layer and the encapsulation layer. However, as the number of TPU layers increases, the stretchability of the circuit will be compromised (Fig. 4h).

We also found that the inclination angle of the e-vests has little effect on the stretchability (Fig. 4i), and we believe that the strain distribution near the

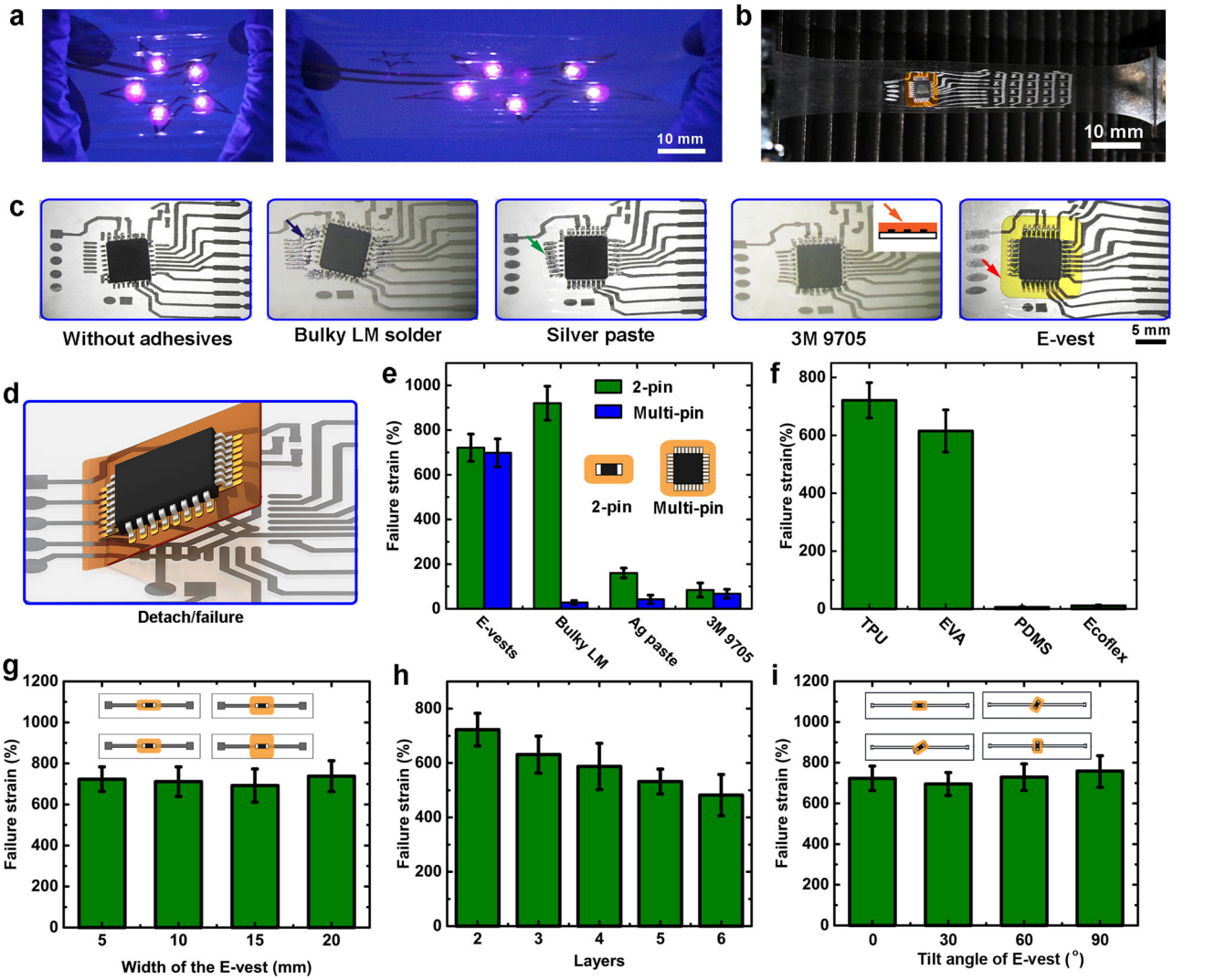


Fig. 4 | Robustness characterization of the soft–rigid connection. a Stretchable circuit with 5 LEDs. **b** A microcontroller chip under tensile strain. **c** Comparison of different strategies to reinforce the soft–rigid connection; Liquid metal solder (black arrow), silver paste (green arrow), 3M 9705 (orange arrow), and e-vests (red arrow). **d** Schematic illustration of a microcontroller chip failing under tensile strain. **e** Comparison of the soft–rigid connection failure strain between 2-pin and multi-pin electronic components. **f** Failure strain of the soft–rigid connection on different substrates. **g** Failure strain of the soft–rigid connection versus width of the E-vest. **h** Failure strain of the soft–rigid connection versus layers of the stretchable circuit. **i** Failure strain of the soft–rigid connection versus the inclination angle of the e-vest. Inset, Schematic showing different inclination angles of the e-vest.

e-vest is similar despite the different inclination angles of the e-vest (Supplementary Fig. 12). Through finite element simulations, we found that an increase in the number of e-vests in the direction of deformation will greatly increase the deformation of the regions between the electronic devices (Supplementary Fig. 13).

Assembly and characterization of stretchable display

In brief, to assemble the stretchable display, we first fabricate each layer in the stretchable display and then assemble it layer by layer via thermal bonding (Fig. 1f).

We can independently fabricate each layer and component of the stretchable display, including two MPC interconnects printed TPU layers, VIA patches, reinforcement films, and e-vest (Fig. 5a). We use a laminator to thermally bond those components to obtain the soft-printed circuit board. Finally, microcontrollers, resistors, and LEDs can be soldered directly to e-vest through solder paste to be a stretchable display (Fig. 5b). LEDs in the stretchable display could be controlled individually to create a scrolling message. The schematic diagram of the circuit design of the two displays is shown in Supplementary Fig. 14. The surface density of electronic components on stretchable circuits reaches 9 devices per square centimeter.

The LED displays are soft and exhibit excellent stretchability. Displays can withstand deformations such as bending, twisting, and stretching (Fig. 5c, d, Supplementary Video 1). The LED displays can conform to irregular surfaces of our skin. In the future, it can be integrated with the skin and interact with people directly on the skin, replacing the functions of smartphones. In the stretching of <100% strain, LEDs in the display are tightly adhered to the substrate and have no tendency to detach.

In order to test the stretch limit of the display, we used a tensile testing machine to evenly stretch the display from its original length until half of the LED failed (Fig. 5e, Supplementary Video 2). We found that our stretchable display could withstand up to 220% strain before losing any LEDs. When we continue to stretch the display, the display will not be damaged immediately, but dead pixels will gradually appear, and the number of failed LEDs will gradually increase as the strain increases. When the tensile strain reaches about 350%, half of the LEDs in the display will fail (Fig. 5f, Supplementary Video 2). We analyzed the causes of LED failure. We found that the TPU

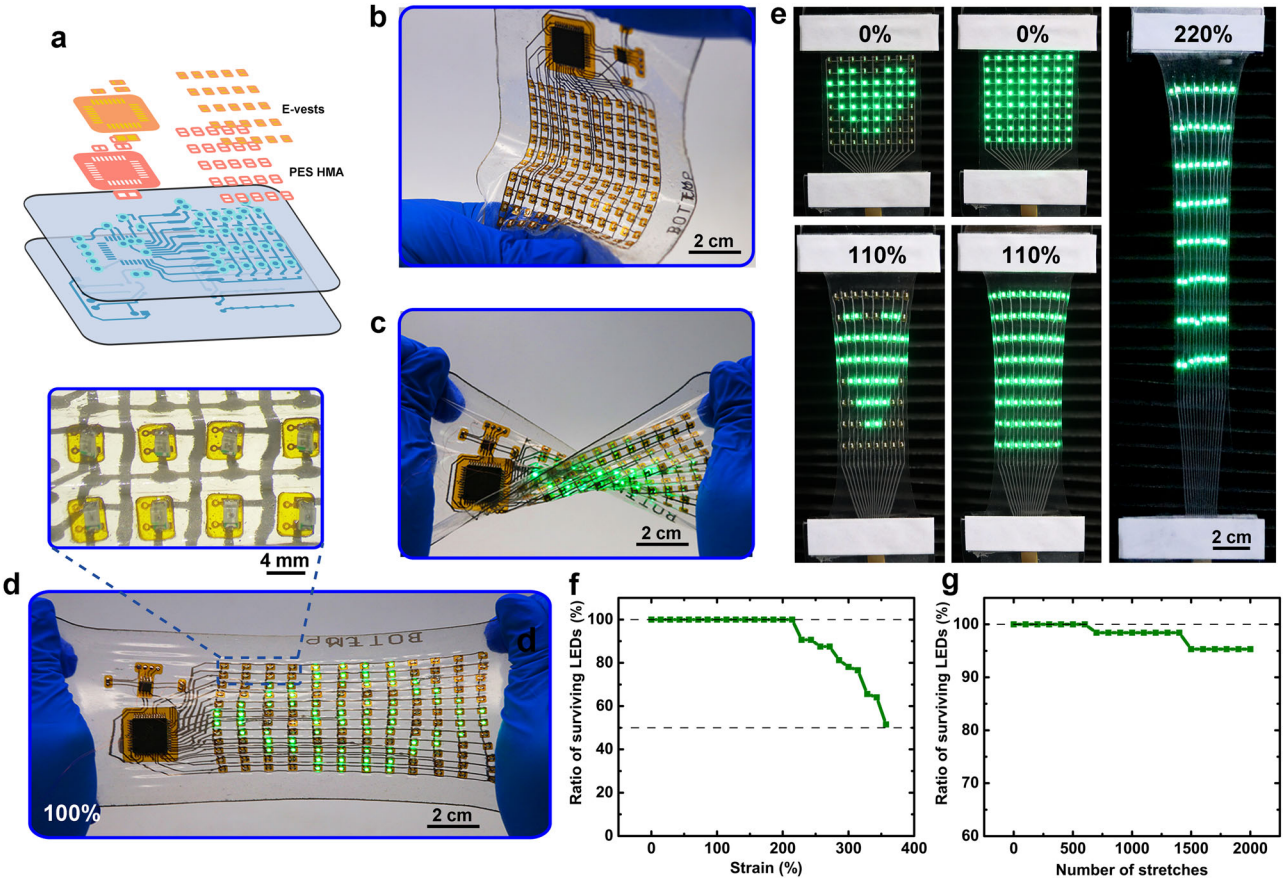


Fig. 5 | Assembly and characterization of stretchable display. a Schematic illustration of different layers and components of the stretchable display. **b** Optical image of the soft 12×12 LED display. **c** The stretchable display under twisting and **d** stretching. **e** The stretchable display was tested under different strains. The display alternately exhibits a heart shape and a square shape. **f** Ratio of surviving LEDs under different tensile strains. **g** Ratio of surviving LEDs after different stretching cycles at 100% strain.

substrate at the edge of the e-vests will break due to the stress concentration, causing the LED to be disconnected from the MPC interconnects (Supplementary Fig. 15). The deformation our display can withstand far exceeds the deformation that human skin can reach.

To test the robustness of the display, we stretched the stretchable display for 2000 cycles at 100% strain. No LEDs failed in the first 500 stretching cycles. After 2000 cycles, the defective pixel rate of the display is <5% (Fig. 5g). Therefore, our stretchable display is fully capable of displaying various information on the skin.

In summary, we achieved ultra-stretchable displays by greatly strengthening soft–rigid connections through electronic vests and reinforcement layers. By adopting the previously developed MPC as interconnects, a circuit containing one rigid electronic component can withstand more than 700% strain, while a display containing hundreds of rigid electronics can withstand 220% strain without dead pixels. Also, the stretchable display exhibits a stable behavior for 500 cycles of 100% strain. Each layer and component in the stretchable display can be fabricated independently, and finally, they can be assembled through thermal bonding. Compared with stretchable displays based on intrinsically stretchable LEDs [38,39], our strategy is more straightforward and less costly. Our display can be assembled by thermal bonding, which does not involve toxic chemicals.

The use of e-vest in this study can greatly enhance the soft–rigid connection, however, the regional extreme deformation on the soft–rigid connection caused by stress concentration still exists, greatly increasing the failure risk of stretchable circuits. Thus, in order to ultimately eliminate the regional extreme deformation in stretchable circuits, intrinsically stretchable LEDs will eventually replace rigid LEDs in our stretchable circuits.

In the future development direction of stretchable display, we believe that there are three optimization directions. The first is to use intrinsically stretchable LEDs with low Young's modulus, and high stretchability, thereby eliminating the soft–rigid interface between the stretchable wire and the LED. The second is to further reduce the size of the stretchable wires and LEDs to increase the resolution of the stretchable display. The third is to increase the current efficiency of LEDs and reduce current loss in wires, ultimately enabling a high-resolution soft display for smartphones that blends in with the skin.

Based on the thermal bonding strategy, the fabrication of multilayer stretchable circuits is even faster and more environmentally friendly than the fabrication of conventional printed circuit board (PCB), which involves polluting and time-consuming processes such as electroplating and chemical etching. With highly stretchable conductors and robust soft–rigid connections, most commercial electronic components can be directly integrated into stretchable circuits, decreasing the urgent need to develop soft alternatives for electronics. Our approach is suitable for converting PCB-based circuits with a large number of electronics and a high level of device integration into stretchable and conformal circuits in a cost-effective manner. Thus, sophisticated functions originally implemented on the PCB have the opportunity to be realized on highly dynamic and irregular surfaces such as human organs and soft robots. Bluetooth chips or Bluetooth modules can also be integrated into our stretchable circuits. In future work, we will integrate Bluetooth chips into our stretchable displays to remotely control the displayed information through mobile phones. At the same time, we also need to study the factors that affect the quality of wireless communications in stretchable circuits [40,41], including communication protocols,

electrical interconnects, power consumption, substrates, and choice of electronic components for stretchable systems.

Methods

Fabrication of the MPC interconnects

The fabrication of the MPC interconnects begins with fabricating the MPC ink based on liquid metals. Briefly, to obtain the MPC ink[35], we added 1 g of PVP [polyvinylpyrrolidone; Mn (number-average molecular weight) = 360,000, Aladdin, China] into 19 g of hexyl alcohol (98%, Macklin, China) and stirred for 48 h to obtain the PVP solution, we sonicate 3 g Gallium Indium eutectic alloy (EGaIn, 75.5% gallium and 24.5% indium by weight, Changsha Kun Yong Materials Co. Ltd., China) in 1 mL PVP solution at 20% amplitude for 60 s using a probe sonicator (S450D, Branson, USA). We used screen-printing (300 mesh stencil) to print the MPC ink on the TPU film (with release film, 3412, 1mil, Bemis, USA) to obtain the MPC wires on the MPC card. After the MPC printing, we dry the MPC wires in an oven (DHG-9420A, Yiheng Scientific Instrument Co. Ltd., China) at 80 °C for 5 min. We use a laser cutter (355 nm Nd:YAG laser, 5 W, Shenzhen JPT Opto-electronics Co., Ltd., China) to drill holes on the MPC interconnects (diameter: 0.2 mm) at the mark speed of 60 mm s^{-1}.

MPC wires are non-conductive after printing because the liquid metal particles within the MPC are covered with an insulating oxide layer. However, the MPC can be mechanically sintered by stretching. We stretched the TPU substrate to a strain of 50% to activate the MPC wires on the substrate. Because the strain applied on the MPC wires can break the insulating oxide layer and form conductive paths between liquid metal particles (Supplementary Fig. 16), the MPC wires will become conductive and stretchable[35,36].

Fabrication of the VIA patches

We use the MPC ink to print circular MPC patterns (diameter: 1 mm) on the TPU film (with release film) and dry them in an oven at 80 °C for 5 min. We use a laser to cut along the outer edge line of MPC at the mark speed of 30 mm s^{-1} to get a rounded rectangular shape. We use a laser cutter to cut along the contour of the MPC with a cutting diameter of 2 mm. The mark speed of the laser is 60 mm s^{-1}, and at this speed, the laser only cuts through the TPU film but not the release film under the TPU. Finally, we peel off the excess TPU film from the release film and leave the VIA patches on the release film (Supplementary Fig. 17).

Fabrication of the reinforcement film

The reinforcement film we used in the stretchable display is the PES film to increase the adhesion between the E-vest and the TPU. We obtain the reinforcement card from commercial PES film (with release film, XJS130, 0.05 mm) by laser ablation at a mark speed of 60 mm s^{-1} (Supplementary Fig. 18). The materials used to screen the reinforcement layer, including the PES, PO (XJO115, 0.05 mm), EVA (HJV90, 0.05 mm), and PA (XJA110, 0.05 mm) are from Shanghai Xingxia Polymer Products Co., Ltd.

Fabrication of the e-vests

The e-vests are customizable commercially available FPC with identical patterns on both sides. The e-vests composed of pPI for different rigid electronics are processed from commercial FPC (Shenzhen Ruixing Express PCB Co. Ltd.). Briefly, during the FPC manufacturing process, a laser is used to drill holes in a polyimide film covered with copper film on both sides. After drilling, copper is electroplated inside the hole wall to obtain the plated through-holes that can achieve interconnection between the copper films on both sides. After steps such as exposure, development, and etching, the unnecessary copper film is removed to obtain an FPC with the desired copper pattern.

The e-vests of all electronic components in the stretchable display are designed in the same FPC, and the position of the e-vest in the FPC is the same as its position in the circuit (Supplementary Fig. 19a). The FPC is then attached to a release paper (0.09 mm, Xianyouyue, PET, low tack), the outlines of different e-vests are cut with a laser, and the excess PI is removed (Supplementary Fig. 19b). Finally all the e-vests can be transferred to the circuit at the same time.

Assembly of the display

The VIA patches on the release film are aligned with the holes on the top MPC layer. After thermal bonding using a laminator (HD 330T, Huanda, China) at a temperature of 140 °C, the TPU on the VIA patch will fuse with the TPU on the top MPC layer and form a mechanical connection. We use another layer of TPU as the encapsulation layer, and use a laser to punch holes in the TPU to expose the contacts connected to the e-vest. The TPU encapsulation layer is then bonded to the top MPC layer via thermal bonding (140 °C). After the encapsulation of the top MPC layer, the bottom MPC layer is aligned with the top MPC layer and also bonded by thermal bonding (140 °C). The PES layer is connected to the top MPC layer through thermal bonding (160 °C), and the e-vests are aligned with the PES and connected to the MPC layer via thermal bonding (140 °C).

We use screen-printing to print solder paste (Sn42Bi58, Ausbond, China) on copper contacts on e-vests (Supplementary Fig. 20). We align the pins of the electronic components with the copper contacts and treat the solder paste in an oven at 150 °C for 5 min. Finally, we finished the fabrication of the stretchable displays.

Characterization

The tensile test and the adhesion force between the PI and different materials (PES, EVA, PA, TPU, PO, Ag paste, and 3M 9705) are measured by the 180° peel test using the universal testing machines (3365, Instron, USA). In the tensile tests, all samples were laser ablated in half-dumbbell shapes (Supplementary Fig. 3) with a width of 10 mm. The two different specimens are connected by thermal bonding. The length of the junction is 10 mm. The initial length of stretching is 50 mm, and the speed of stretching is 50 mm/min. The 180° peel test is shown in Fig. 2a. The length and the width of the junctions are 10 mm, and the speed of stretching is also 50 mm/min. We used the programmable linear guide slide (FSL_40, FUYU, China) to stretch samples for the stretch test at the speed of 100 mm/min and the cyclic test at the speed of 1000 mm/min. We used an electrochemical station (1040C, CHI, China) to record the electrical resistance in real-time during tensile tests and cycle tests. The surface morphology of the MPC is characterized by the scanning electron microscope (SEM, ZEISS Merlin). Optical micrographs were obtained from a Stereo microscope (SZM-45B2, Gaopin, China). The contact angle of different polymers (PES, EVA, PA, TPU, and PO) to polyimide is achieved by melting those polymers at 160 °C in a hot plate and observing the contact angle through the dissecting microscope. The stretchable LED displays are tested in the hands of authors (obtained informed consent from all participants) and approved by the institutional ethics committee (Medical Ethics Committee of Capital Medical University, 2021SY021).

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

L.T. and X.J. conceived and supervised this project. L.T. designed, fabricated, and characterized the E-cards. J.R. performed finite element simulations. L.T., H.W., and X.J. wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
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